先进计算与数字工程研究中心

PATTERN RECOGNITION AND MACHINE LEARNING

Lecture 04 – Discriminative Classifiers I: Linear, Generalized, Piecewise, and Fisher



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A tomato reaches the sorting gate

The observation

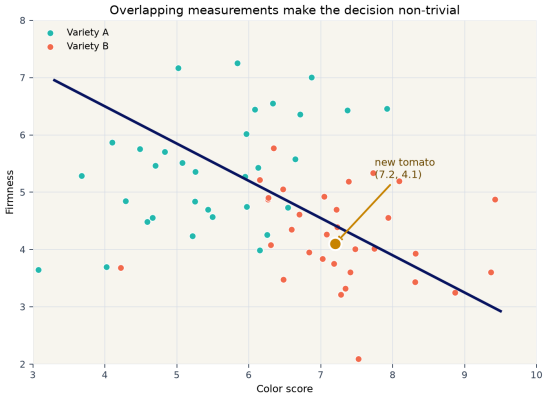
Color score $x_1 = 7.2$ and firmness $x_2 = 4.1$.

The decision

Pack now or hold for another day?

The complication

Two varieties overlap in the sensor space.



The same data support two questions

Question	Quantity	Typical use
How could this measurement arise?	$p(x \mid y)$ and $p(y)$	Describe each class
Which label fits this measurement?	$p(y \mid x)$	Predict a class
Which side of the boundary is it on?	$g(x)$	Make a decision

One data table. Different questions lead to different models.

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Lecture map

Syllabus topic	Question answered
Classifier families	What should the model represent?
Linear discriminant functions	What does one straight boundary mean?
Generalized linear functions	How can a feature map create a curve?
Piecewise linear functions	How can several local lines form one rule?
Fisher linear discriminant	Which projection separates the classes?

Programming

NumPy examples will calculate scores, maps, prototypes, and Fisher directions.

World model and decision model

Generative route

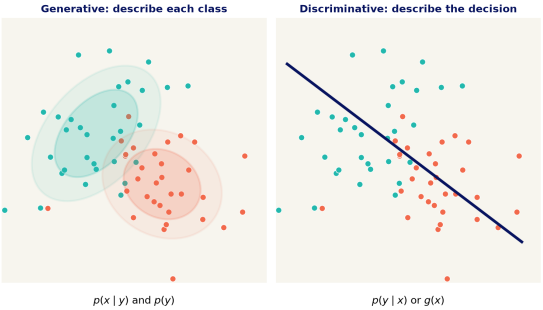
Fit $p(x \mid y)$ and $p(y)$, then derive $p(y \mid x)$.

Discriminative route

Fit $p(y \mid x)$ or a score $g(x)$ directly.

Do not rank them by name

The useful choice depends on data, assumptions, and the required output.



Generative modelling describes each class

Tomato example

Fit a color and firmness distribution for each maturity class.

$$p(y = k \mid x) = \frac{p(x \mid y = k)p(y = k)}{\sum_j p(x \mid y = j)p(y = j)}$$

What the model can do. Classify, sample plausible measurements, and reveal unusual observations.

Discriminative modelling targets the label

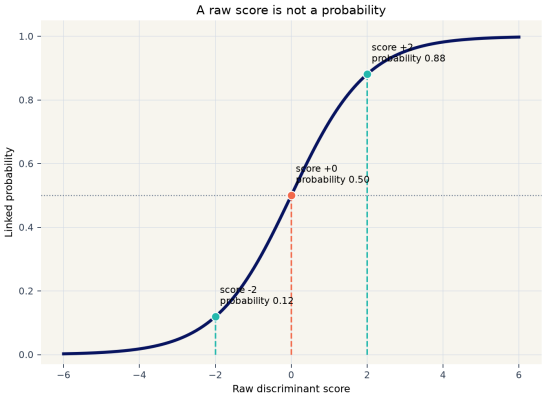
Conditional probability

Estimate $p(y \mid x)$ when calibrated uncertainty matters.

Decision score

Estimate $g(x)$ when only the class decision is required.

$$\hat{y} = \arg\max_k p(y = k \mid x) \quad \text{or} \quad \hat{y} = \operatorname{sign} g(x)$$



Bayes can produce a discriminant function

Start from the posterior

Choose class 1 when $p(y = 1 \mid x) > p(y = 2 \mid x)$.

$$g(x) = \log p(x \mid y = 1) + \log p(y = 1) - \log p(x \mid y = 2) - \log p(y = 2)$$

Same decision

Class 1 is selected when $g(x) > 0$.

What unlabeled inputs can contribute

Possible generative use

Unlabeled x values can help estimate the structure of $p(x)$ under a stated model.

No automatic benefit

A wrong mixture or cluster assumption can move the boundary in the wrong direction.

Purely supervised discriminative fit

Without an added objective, unlabeled rows contribute no label loss.

Fewer assumptions can focus capacity on the boundary

A direct target

A discriminative model need not reproduce every detail of $p(x)$.

Tomato sensor

Lighting variation that does not change the label may be ignored by the boundary.

The tradeoff

Less structure also means fewer ways to use missing labels or detect implausible inputs.

When generative assumptions help

Situation	Why the structure helps
Few labels and credible class models	Parameters share information through the model
Missing measurements	The joint distribution can marginalize missing entries
Anomaly screening	Low likelihood can flag an unfamiliar input
Simulation is required	The fitted model can produce synthetic observations

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When generative assumptions hurt

Misspecified shape

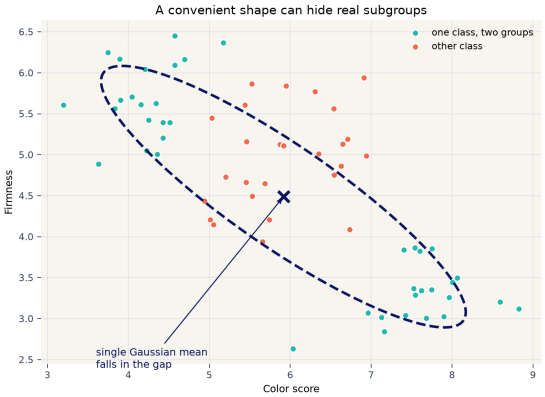
A Gaussian ellipse cannot represent every class distribution.

Hidden groups

Two tomato varieties may create two modes inside the same maturity class.

Consequence

The estimated likelihood can be wrong far from the boundary and still alter the decision.



A fair comparison uses the same evidence

1. Use the same training, validation, and test split.
2. Tune both model families on the validation set.
3. Evaluate the metric that matches the decision cost.
4. Inspect calibration only when probabilities are claimed.

Common mistake

Comparing one tuned model with another model's default settings proves little.

Case change under a different light

Morning

Color and firmness separate the two decisions with a simple diagonal boundary.

Afternoon

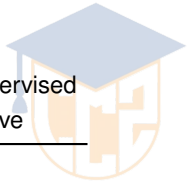
A warm lamp shifts the color score by $+1.5$ for both classes.

Question

Which part should change: the data model, the features, or the boundary bias?

Classifier family choice

Need	First model to consider
A direct boundary with many labels	Discriminative
A class distribution and simulated inputs	Generative
Useful unlabeled structure with credible assumptions	Generative or semi supervised
Only calibrated class probabilities	Conditional discriminative



Thirty second answer

Choose one row and state the assumption that makes your choice reasonable.

Four questions for every boundary

1. What representation enters $g(x)$?
2. What shape can $g(x) = 0$ express?
3. How does the score become a class decision?
4. What evidence would reveal a bad assumption?

Use this checklist

Apply the same four questions to linear, mapped, piecewise, and Fisher boundaries.

Linear discriminant functions

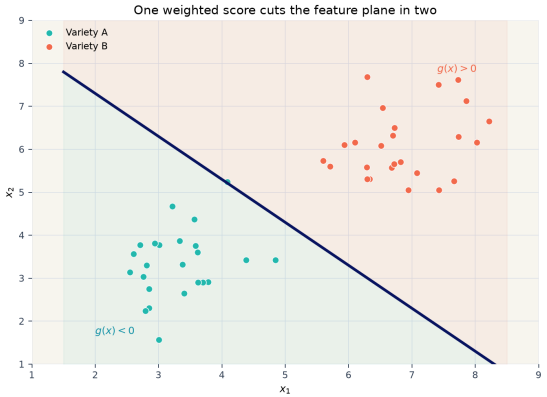
The model

A weighted sum turns a feature vector into one score.

$$g(x) = w^\top x + w_0$$

The boundary

All points satisfying $g(x) = 0$.



A score turns measurements into a decision

Two class rule

Predict class $+1$ when $g(x) > 0$. Predict class -1 when $g(x) < 0$.

At exactly zero

Use a fixed tie rule or send the sample for review.

A score is not a label

The sign creates the label. The magnitude describes confidence only under extra assumptions.

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Two features make a line

$$g(x) = w_1 x_1 + w_2 x_2 + w_0$$

Decision boundary

$w_1 x_1 + w_2 x_2 + w_0 = 0$

If $w_2 \neq 0$

$$x_2 = -\frac{w_1}{w_2} x_1 - \frac{w_0}{w_2}$$

Geometric meaning. Slope $-w_1/w_2$ and vertical intercept $-w_0/w_2$.

The weight vector is normal to the boundary

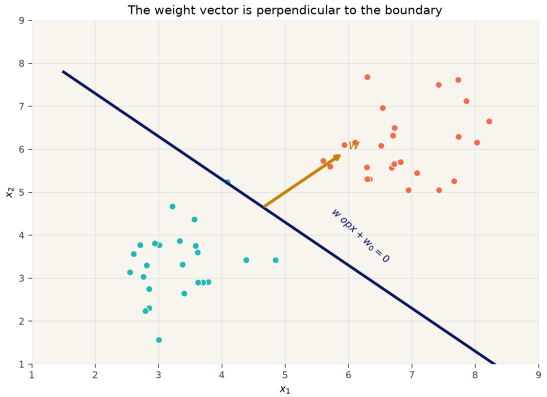
A movement along the boundary

For boundary points x_a and x_b, $w^\top (x_a - x_b) = 0$.

$$w \perp (x_a - x_b)$$

Direction of increase

Moving in the direction w increases $g(x)$ fastest.



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The bias moves the line

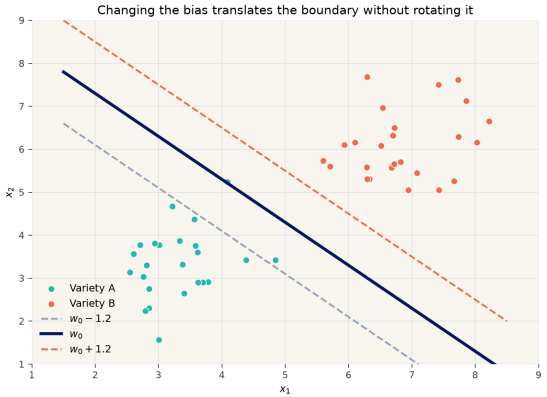
Keep w fixed

Changing w_0 moves the boundary parallel to itself.

$$\text{shift along } w = -\frac{\Delta w_0}{\|w\|}$$

Sensor recalibration

A constant color offset can often be absorbed by the bias.



Signed score and geometric distance

Raw score

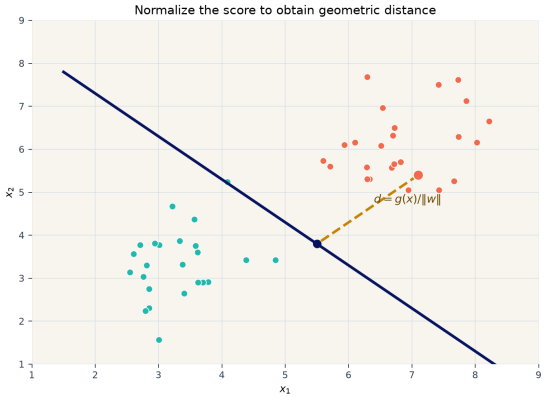
$g(x) = w^\top x + w_0$ changes if every parameter is rescaled.

Signed distance

$$d(x) = \frac{w^\top x + w_0}{\|w\|}$$

Why divide

$g(x)$ and $10g(x)$ define the same boundary but not the same raw magnitude.



A calculation at the sorting gate

Teaching parameters

$w = (1, -0.8)^\top$, $w_0 = -3.5$, $x = (7.2, 4.1)^\top$

$$g(x) = 7.2 - 0.8 \times 4.1 - 3.5 = 0.42$$

Decision

The score is positive, so the model selects pack now.

Distance. $0.42/\sqrt{1^2 + 0.8^2} \approx 0.328$ feature units.

Feature units change the weights

Two equivalent records

Firmness 4.1 on a 0 to 10 scale equals 41 on a 0 to 100 scale.

Equivalent model

The weight -0.8 becomes -0.08 when that feature is multiplied by 10.

Wrong shortcut

Keeping the old weight after changing units moves the boundary.

Standardization makes scales comparable

Transform each feature

$$z_j = \frac{x_j - \mu_j}{s_j}$$

Interpretation

A one unit change in z_j means one training standard deviation.

Deployment rule

Reuse the training values of μ_j and s_j for new samples.

The augmented vector absorbs the bias

Augmented feature

$\tilde{x} = (x_1, \ldots, x_D, 1)^\top$

Augmented weight

$\tilde{w} = (w_1, \ldots, w_D, w_0)^\top$

$$g(x) = \tilde{w}^\top \tilde{x}$$

Why it helps

One inner product handles both feature weights and the bias.

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Equivalent parameters and one boundary

Positive rescaling

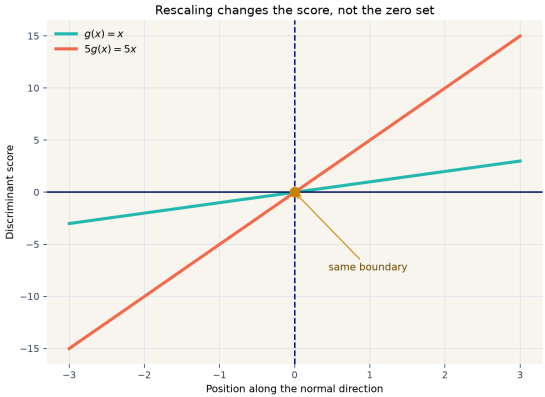
(w, w_0) and (cw, cw_0) define the same decision for every $c > 0$.

Negative rescaling

For $c < 0$, the boundary stays but the class labels swap.

Identifiability

A normalization or training objective selects one representative from many equivalent parameter sets.



Linear separability

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Separable training set

There exists (w, w_0) such that $y_i(w^\top x_i + w_0) > 0$ for every sample.

Geometric view

One hyperplane places every positive sample on one side and every negative sample on the other.

Training property

Separable training points do not guarantee perfect predictions on future data.

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Exclusive or cannot use one straight boundary

x_1	x_2	Positive class
0	0	Yes
0	1	No
1	0	No
1	1	Yes

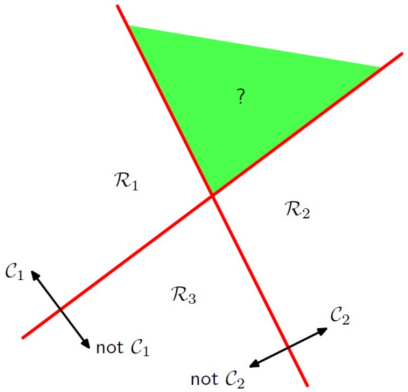
Contradiction

No affine function can place both diagonal corners on one side and the other corners on the opposite side.

Later solutions

Add an interaction feature, combine local lines, or use a deeper model.

One-versus-rest classification



One-versus-rest scores can disagree in overlapping regions; define how to choose the final class.

Ambiguous regions in one versus rest

More than one positive score

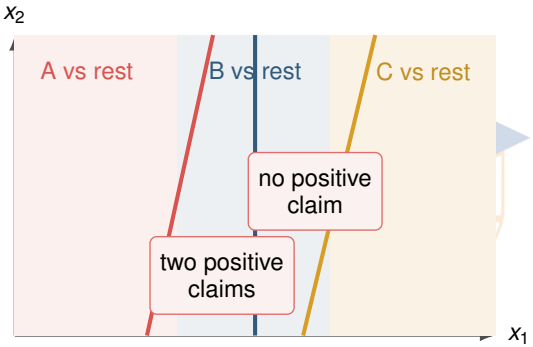
Several classes claim the same sample.

No positive score

No classifier claims the sample.

Required policy

Use comparable scores, a priority rule, or a reject option. State the choice explicitly.



Independent binary rules need a policy for overlaps and gaps.

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One versus one classification

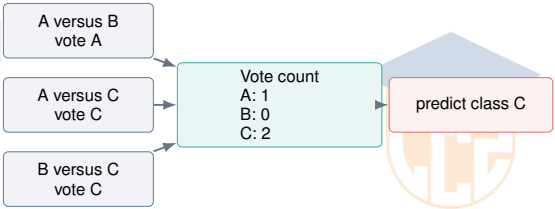
Pairwise rule

Fit one classifier for every pair of classes.

$$\text{number of classifiers} = \frac{C(C-1)}{2}$$

Prediction

Each pair casts one vote. The class with most votes wins.



$C(C-1)/2$ pairwise classifiers; ties need a declared rule.

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Pairwise voting can tie

Three pairwise results

A beats B, B beats C, and C beats A.

Class	Votes
A	1
B	1
C	1

A complete classifier needs a tie rule

Examples include summed margins, calibrated pairwise probabilities, or a fixed class order.

One discriminant score per class

Shared comparison

Fit C scores and choose the largest.

$$\hat{y} = \arg\max_c g_c(x), \qquad g_c(x) = w_c^\top x + w_{c0}$$

Pairwise boundary

Classes c and k meet where $g_c(x) = g_k(x)$.



Argmax partitions the whole plane

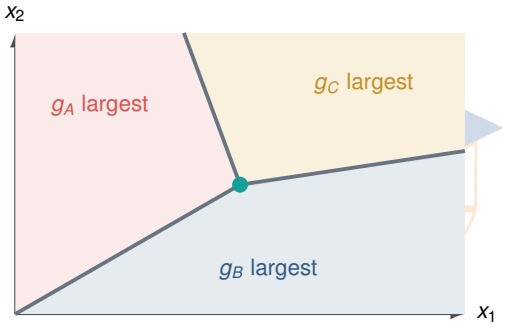
Complete assignment

Every point has at least one largest score.

Ties

Two or more classes can share the maximum exactly on a boundary.

Geometry. The visible class regions are intersections of linear inequalities.



Every point receives the class with the largest score; ties lie on boundaries.

A numeric three class decision

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Scores

$g_1 = -x_1 + x_2$, $g_2 = x_1 + x_2 - 5$, $g_3 = -x_2 + 1$

Input	g_1	g_2	g_3	Argmax
$(6, 5)$	-1	6	-4	Class 2
$(3, 5)$	2	3	-4	Class 2



One versus rest difference

At $(3, 5)$, both g_1 and g_2 are positive.

Classifier count grows differently

Classes	One versus rest	One versus one	Comparable class scores
3	3	3	3
10	10	45	10
100	100	4,950	100



Cost beyond training

Prediction must also evaluate the required scores or pairwise classifiers.

Scores and probabilities answer different questions

Output	What it supports	What it does not guarantee
Score $g(x)$	Ranking and boundary decisions	Calibrated probability
Margin distance	Geometric separation	Frequency among similar cases
Probability $p(y \mid x)$	Expected risk and thresholds	Correctness without validation

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A link function can convert a score

Logistic example

$$p(y = 1 \mid x) = \frac{1}{1 + \exp[-g(x)]}$$

Boundary

$g(x) = 0$ corresponds to probability 0.5.

Preview only

The link defines a probability model. Its fitting and evaluation appear in the linear models lecture.

Least squares can fit a discriminant

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Targets

Encode classes as $t_i \in \{-1, +1\}$.

Objective

$$\min_{\tilde{w}} \|\tilde{X}\tilde{w} - t\|_2^2$$

Closed form when appropriate

Solve $\tilde{X}^\top \tilde{X}\tilde{w} = \tilde{X}^\top t$.

Least squares uses every residual

Correct but extreme

A positive sample with target $+1$ and score $+8$ has residual 7.

Wrong but near

A positive sample with score -0.1 has residual -1.1.

Mismatch

The first point is correctly classified but contributes a much larger squared error.

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A large residual can pull the boundary

Before the extreme point

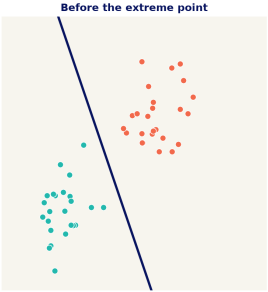
The fitted line follows the main cloud.

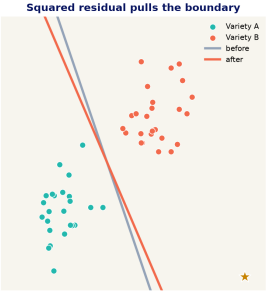
After the extreme point

Squared loss gives the distant residual disproportionate influence.

Response

Inspect leverage, measurement errors, and a classification specific loss.





NumPy batch scoring

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```
import numpy as np

X = np.array([[7.2, 4.1],
              [5.4, 6.0],
              [6.8, 3.5]])
w = np.array([1.0, -0.8])
b = -3.5

scores = X @ w + b
labels = np.where(scores > 0, 1, -1)
print(scores)
print(labels)
```

Expected scores

$[0.42, -2.90, 0.50]$ and labels $[1, -1, 1]$.

NumPy multiclass scores

```
W = np.array([[-1.,  1.],
              [ 1.,  1.],
              [ 0., -1.]])
b = np.array([0., -5., 1.])
X = np.array([[6., 5.], [3., 5.]])

scores = X @ W.T + b
prediction = scores.argmax(axis=1) + 1
print(scores)
print(prediction)
```

Axis check

Rows are samples and columns are class scores. Take the maximum across columns.

Generalized linear discriminant functions

Main idea

Transform x into features $\phi(x)$, then use a linear score in the transformed space.

$$g(x) = w^\top \phi(x)$$

Result

A hyperplane in feature space can appear curved in the original space.

Linear in parameters can curve in input space

Linear parameters

$g(x) = w_0 + w_1\phi_1(x) + \cdots + w_M\phi_M(x)$

Nonlinear feature

If $\phi_1(x) = x^2$, the score is nonlinear in x.

Terminology

Generalized linear here means a fixed feature map followed by a linear discriminant. It is distinct from the statistical family called generalized linear models.

One threshold cannot isolate a middle interval

One dimensional classes

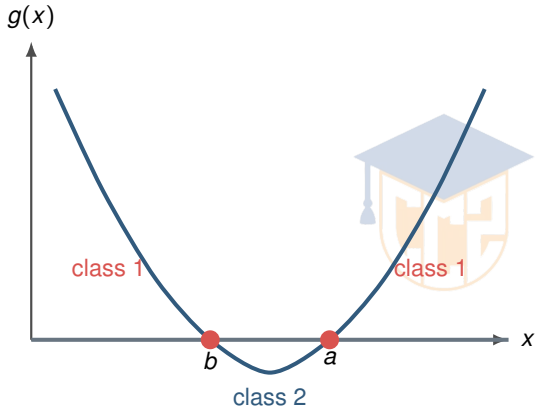
Class 2 occupies $b < x < a$. Class 1 occupies $x \leq b$ or $x \geq a$.

Linear limitation

A one dimensional affine score changes sign at most once.

Needed boundary

The desired rule needs two crossing points.



Two zero crossings isolate the middle interval.

Lift the input into two features

Feature map

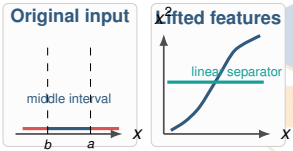
$\phi(x) = (x^2, x, 1)^\top$

$$g(x) = (x - a)(x - b) = x^2 - (a + b)x + ab$$

Linear weights

$w = (1, -(a + b), ab)^\top$

Decision. Positive outside the interval and negative inside when $a > b$.



The quadratic map turns two crossings on the line into one linear separator in feature space.

A line in feature space becomes a parabola

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Transformed coordinates

Let $z_1 = x^2$ and $z_2 = x$.

$$z_1 - (a+b)z_2 + ab = 0$$

Two views

This is a line in the z plane and a quadratic equation on the original x axis.

Exclusive or becomes linear with one interaction

x_1	x_2	$x_1 x_2$	Target y
-1	-1	+1	+1
-1	+1	-1	-1
+1	-1	-1	-1
+1	+1	+1	+1

A one feature rule

$g(x) = x_1 x_2$ and $\hat{y} = \operatorname{sign} g(x)$.

What changed

The interaction records whether the two signs agree.

A quadratic map in two dimensions

Full degree two map

$\phi(x) = (x_1^2, x_1 x_2, x_2^2, x_1, x_2, 1)^\top$

$$g(x) = w_{11} x_1^2 + w_{12} x_1 x_2 + w_{22} x_2^2 + w_1 x_1 + w_2 x_2 + w_0$$

Possible boundaries

Ellipses, parabolas, hyperbolas, pairs of lines, and degenerate cases.

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Quadratic feature count

Component	Number
Quadratic terms	$D(D+1)/2$
Linear terms	D
Bias	1
Total	$(D+1)(D+2)/2$

For two inputs

$3 + 2 + 1 = 6$ features.

For one hundred inputs

5,151 features.

Polynomial degree grows the feature space

All monomials up to degree r

$$M = \binom{D+r}{r}$$

Input dimension D	Degree r	Features M
10	2	66
10	3	286
100	2	5,151
100	3	176,851

Capacity warning

Higher degree can overwhelm both memory and the available sample size.

Powers amplify scale differences

Two raw features

Color ranges near 10 while acidity ranges near 0.1.

After squaring

The scales become about 100 and 0.01.

Numerical effect

Optimization and regularization can become dominated by feature units.

Usual response. Standardize suitable inputs before constructing polynomial terms.

Interactions need a domain meaning

Tomato example

$x_1 x_2$ lets the effect of color depend on firmness.

Interpretation

Deep color may support pack now only when the tomato is not unusually hard.

Spurious interaction

A large coefficient can reflect chance correlation when data are scarce.

A richer map can also fit noise

Training fit

A high degree polynomial can pass close to many training points.

Between the points

The boundary may oscillate in regions with little evidence.

Model selection

Use validation performance, regularization, and a degree chosen before looking at the test set.

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Basis choice comes before weight fitting

Basis	Useful when	Main caution
Polynomial	Smooth global curvature	Feature count grows quickly
Radial basis	Local similarity matters	Centers and widths must be chosen
Spline	Smooth pieces along an ordered variable	Knots control flexibility
Domain interaction	A known mechanism combines features	Interpretation depends on preprocessing

Polynomial features by hand

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```
import numpy as np

X = np.array([[1., 2.],
              [3., 4.]])
x1, x2 = X[:, 0], X[:, 1]
Phi = np.column_stack([
    x1**2, x1*x2, x2**2, x1, x2, np.ones(len(X))
])
print(Phi)
print(Phi.shape)
```

Shape

Two samples become two rows with six quadratic features.

Interval classifier in code

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```
a, b = 3.0, 1.0
x = np.array([0.0, 1.5, 2.0, 3.5])
Phi = np.column_stack([x**2, x, np.ones(len(x))])
w = np.array([1.0, -(a + b), a * b])

scores = Phi @ w
labels = np.where(scores >= 0, 1, 2)
print(scores)
print(labels)
```

Expected labels

Class 1, class 2, class 2, class 1.

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Piecewise linear discriminant functions

Main idea

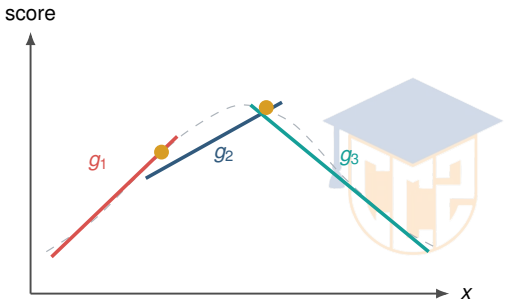
Use several affine pieces, each responsible for a local part of the input space.

Boundary

Straight segments join to approximate a curved or multi region decision surface.

Capacity

The number and placement of pieces must be learned or chosen.



Local affine pieces approximate a curved score while keeping each piece simple.

Several local lines fit a multimodal class

Two tomato varieties

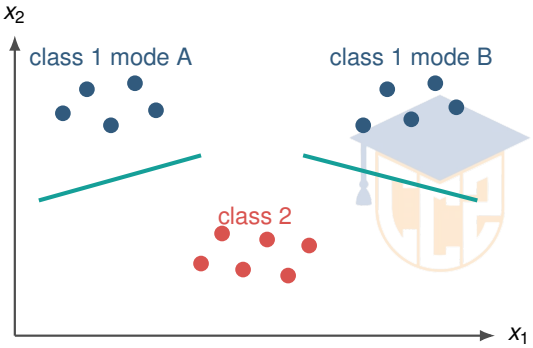
The pack now class forms two separated clusters because the varieties mature at different colors.

One global line

It may cut through the hold cluster or miss one pack cluster.

Local pieces

Each cluster can receive a nearby linear comparison.



Two local separators handle a class with two separated modes.

Three boundary families on one problem

Boundary family	Strength	Main limitation
One affine boundary	Simple and stable	Cannot bend
Feature mapped boundary	Smooth global curvature	Feature count may grow
Piecewise affine boundary	Local flexibility	Pieces and joins need control

No universal winner

Choose the smallest family that captures repeatable structure.

Local rules need a combination rule

Piece scores

$g_{c\ell}(x) = w_{c\ell}^{\top} x + b_{c\ell}$ for piece ℓ of class c.

Class score

$$g_c(x) = \max_{\ell=1,\dots,L_c} g_{c\ell}(x)$$

Prediction

$\hat{y} = \arg\max_c g_c(x)$

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Two affine pieces meet on a line

Switching set

$g_{c1}(x) = g_{c2}(x)$

$$(w_{c1} - w_{c2})^\top x + (b_{c1} - b_{c2}) = 0$$

Meaning

The active piece changes across this hyperplane.

Not always a class boundary

Both pieces may belong to the same class.

Nearest prototypes create affine comparisons

Choose the nearest prototype

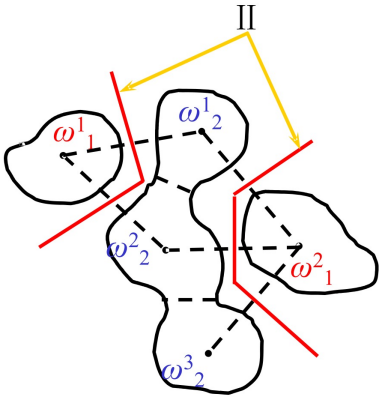
$\hat{j} = \arg\min_j \|x - \mu_j\|^2$

$$-\|x - \mu_j\|^2 = -x^\top x + 2\mu_j^\top x - \|\mu_j\|^2$$

Shared term

$-x^\top x$ cancels when prototypes are compared.

Several prototypes create a piecewise boundary



Several local prototype comparisons form a piecewise boundary.

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Build a piecewise boundary from local linear decisions

- One linear score supplies one straight boundary.
- Combine several local decisions to separate different parts of the space.
- The combined boundary may bend even when each local boundary is straight.
- Check every region: overlapping or uncovered regions need an explicit rule.

The number of pieces controls capacity

Pieces	Likely behavior
1	A single affine boundary
A few	Captures broad local structure
Many	Tracks fine variations and noise
One per sample	Approaches nearest neighbor behavior

Selection

Choose the piece count with validation evidence, not training error alone.

Too few pieces underfit

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Visible symptom

Systematic errors occur in the same region for many samples.

Interpretation

The local class structure is missing from the model.

Useful response. Add a justified prototype or feature, then recheck on validation data.

Too many pieces memorize gaps

Visible symptom

The boundary zigzags between neighboring training points.

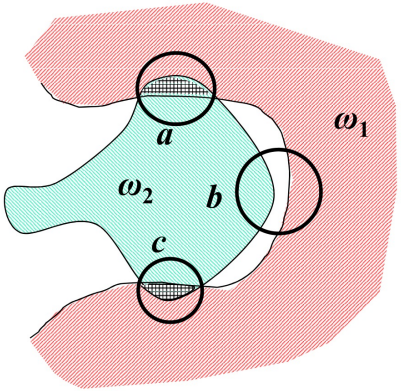
Why it happens

Each small region can react to sampling noise or a mislabeled point.

Useful response

Merge prototypes, require minimum support, or penalize model complexity.

Train near overlapping classes



Local comparisons are most difficult where the class regions meet.

Prototype pairs suggest initial local planes

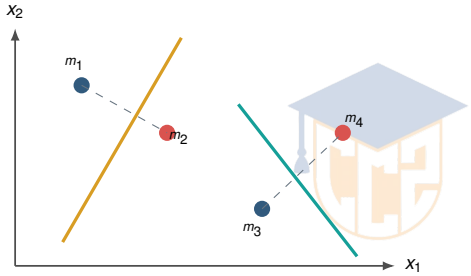
Mutual nearest prototypes

A prototype from each class is the other's nearest opposite class prototype.

$$\left[x - \frac{\mu_a + \mu_b}{2}\right]^\top (\mu_a - \mu_b) = 0$$

Initial plane

Use the perpendicular bisector, then refine with nearby samples.



A perpendicular bisector gives an initial local plane for each opposite-class prototype pair.

Nearest prototype classification in NumPy

```
prototypes = np.array([[7.0, 4.0], [5.5, 2.5],
                       [4.0, 6.0], [2.5, 4.5]])
prototype_labels = np.array([1, 1, 2, 2])
X = np.array([[6.8, 3.7], [3.0, 5.2], [5.0, 4.2]])

dist2 = ((X[:, None, :] - prototypes[None, :, :])**2).sum(axis=2)
nearest = dist2.argmin(axis=1)
prediction = prototype_labels[nearest]
print(dist2)
print(prediction)
```

Array shape

The distance matrix has three samples by four prototypes.

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Fisher linear discriminant

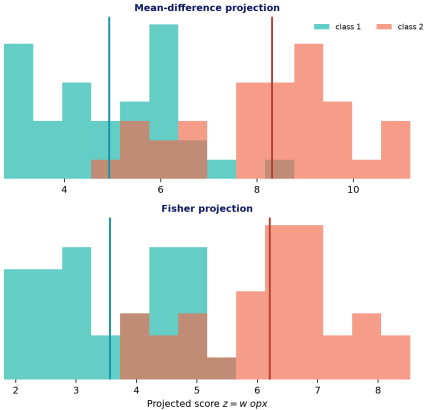
Goal

Project many features onto one axis while preserving class separation.

$$z = w^\top x$$

Fisher criterion

Projected class centers should be far apart and each class should remain compact.



Projection can preserve or destroy separation

Same points

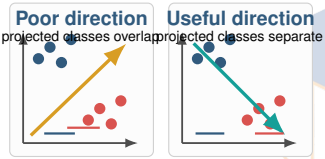
Only the direction of projection changes.

Poor direction

Different classes land on overlapping positions.

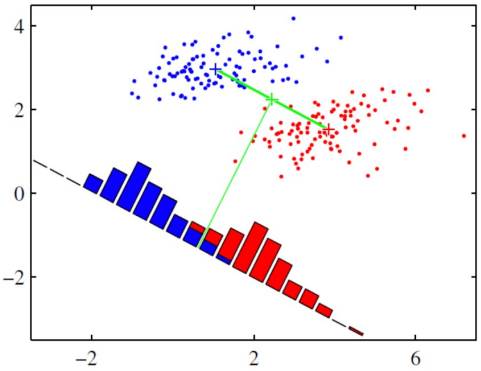
Useful direction

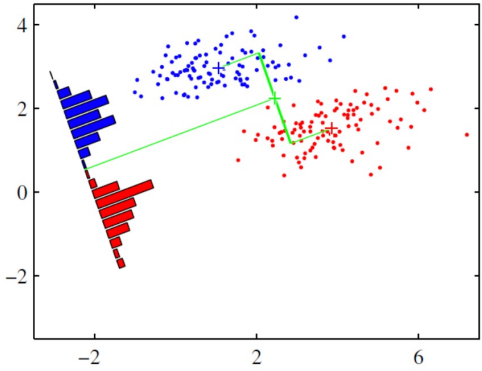
Class centers separate and within class spread stays small.



Projection changes the one-dimensional overlap, not the original points.

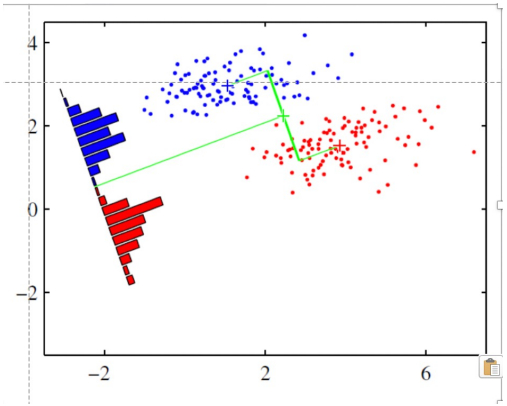
Compare two projection directions





Compare class overlap after projection, not only the appearance of the original point cloud.

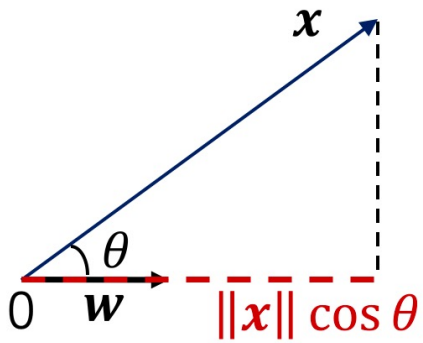
Fisher separates means relative to within-class spread



A useful direction separates class means relative to within-class spread.

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Projection is a dot product



For a unit direction w, the projected coordinate is $w^\top x = \|x\| \cos\theta$.

Projected class means

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Original mean

$$m_k = \frac{1}{N_k} \sum_{i:y_i=k} x_i$$

Projected mean

$$\tilde{m}_k = \frac{1}{N_k} \sum_{i:y_i=k} w^\top x_i = w^\top m_k$$

Consequence

Projected center difference is $w^\top (m_1 - m_2)$.

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Projected within class spread

Class scatter

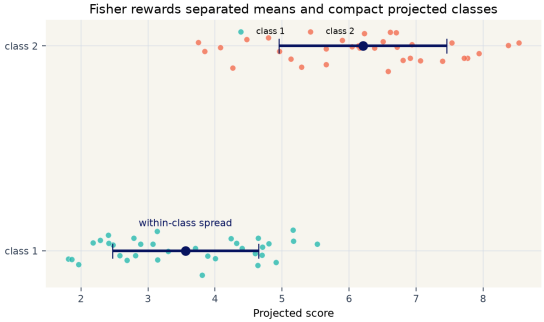
$$S_k = \sum_{i:y_i=k} (x_i - m_k)(x_i - m_k)^\top$$

Projected scatter

$$\tilde{s}_k^2 = \sum_{i:y_i=k} (w^\top x_i - w^\top m_k)^2 = w^\top S_k w$$

Total within scatter

$w^\top S_W w$ where $S_W = S_1 + S_2$.



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Between class separation

Scalar separation

$$(\tilde{m}_1 - \tilde{m}_2)^2 = [w^\top (m_1 - m_2)]^2$$

Matrix form

$$S_B = (m_1 - m_2)(m_1 - m_2)^\top$$

$$[w^\top (m_1 - m_2)]^2 = w^\top S_B w$$

The Fisher criterion is a ratio

Large numerator

Projected class means should be far apart.

Small denominator

Projected samples within each class should stay compact.

$$J(w) = \frac{w^\top S_B w}{w^\top S_W w} = \frac{[w^\top (m_1 - m_2)]^2}{w^\top S_W w}$$

The Fisher ratio ignores scale

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Rescale the direction

Let $w' = cw$ for any nonzero c.

$$J(cw) = \frac{c^2 w^\top S_B w}{c^2 w^\top S_W w} = J(w)$$

Practical normalization

Use $w/\|w\|$ for display, or fix $w^\top S_W w = 1$ in a derivation.

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The mean difference can point the wrong way

Naive direction

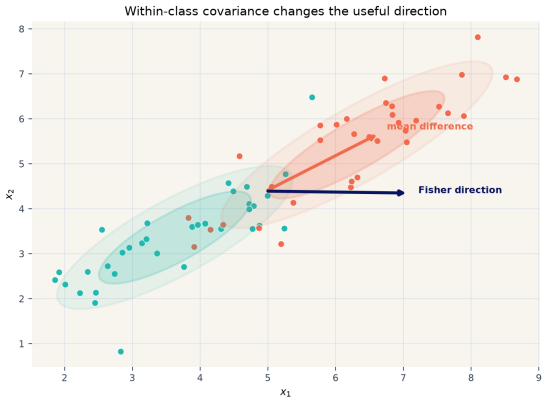
$w = m_1 - m_2$

Problem

It ignores directions where each class has large variance.

Elongated clouds

The center line may align with a noisy long axis and create overlapping projections.



Deriving the Fisher direction

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Constrained form

Maximize $w^\top S_B w$ subject to $w^\top S_W w = 1$.

$$S_B w = \lambda S_W w$$

Two class simplification. $S_B w = (m_1 - m_2)(m_1 - m_2)^\top w$ is a scalar multiple of $m_1 - m_2$.

Within class scatter changes the direction

Fisher direction

$$w \propto S_W^{-1}(m_1 - m_2)$$

High variance direction

The inverse scatter reduces its contribution.

Low variance direction

A modest mean difference can matter when measurements are consistent.

Scatter matrices come from centered samples

1. Compute m_1 and m_2 from their own classes.
2. Center every sample using its class mean.
3. Accumulate outer products inside each class.
4. Add the two matrices to obtain S_W.

Common mistake

Centering all samples by the global mean mixes between class separation into the within class scatter.

Solve a linear system instead of an inverse

Mathematical expression

$w = S_W^{-1}(m_1 - m_2)$

Numerical computation

Solve $S_W w = m_1 - m_2$.

```
Sw = np.array([[4., 1.], [1., 2.]])
mean1 = np.array([1., 1.])
mean2 = np.array([3., 2.])
w = np.linalg.solve(Sw, mean1 - mean2)
```

Reason

Explicit matrix inversion is slower and usually less numerically stable.

A two dimensional Fisher example

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Class means

$m_1 = (2, 0)^\top$ and $m_2 = (2, 2)^\top$

Within class scatters

$$S_1 = \begin{bmatrix} 1 & 1/2 \\ 1/2 & 1 \end{bmatrix} \text{ and } S_2 = \begin{bmatrix} 1 & -1/2 \\ -1/2 & 1 \end{bmatrix}$$

$$S_W = 2I, \qquad w = S_W^{-1}(m_1 - m_2) = (0, -1)^\top$$

Projected scores in the numeric example

Point	$z = w^\top x$	Side of midpoint
$(2, 0)$	0	Class 1 center
$(2, 2)$	-2	Class 2 center
$(4, 0.4)$	-0.4	Class 1 side
$(1, 1.6)$	-1.6	Class 2 side

Midpoint threshold

$t = (0 - 2)/2 = -1$.

A threshold is chosen after the direction

Projection

Fisher selects w to separate one dimensional class distributions.

Classification

Choose a threshold t and compare $z = w^\top x$ with t.

Equal symmetric case

The midpoint $(w^\top m_1 + w^\top m_2)/2$ is a common choice.

Separate decisions

The Fisher criterion alone does not encode priors or unequal error costs.

Class priors move the projected threshold

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Equal projected variance

If $z \mid k \sim \mathcal{N}(\mu_k, \sigma^2)$, Bayes gives

$$t = \frac{\mu_1 + \mu_2}{2} - \frac{\sigma^2}{\mu_1 - \mu_2} \log \frac{\pi_1}{\pi_2}$$

Interpretation

A more common class receives a larger region along the projection axis.

Decision costs can move the threshold again

Asymmetric consequence

Packing an unripe tomato may cost more than holding a ripe one for a day.

Risk rule

Choose the action with the smaller conditional expected loss.

What remains fixed

Changing the cost changes the action threshold, not the fitted class distributions.

Fisher and equal covariance Gaussian discrimination

Shared covariance model

$x \mid y = k \sim \mathcal{N}(\mu_k, \Sigma)$

$$w_{\text{Gaussian}} = \Sigma^{-1}(\mu_1 - \mu_2)$$

Direction

Fisher uses $S_W^{-1}(m_1 - m_2)$, which has the same form.

Threshold

Priors and losses still determine the intercept of the Gaussian decision rule.

High dimensional scatter can be singular

Why singularity occurs

Features may be duplicated, constant, or more numerous than independent centered samples.

Image data

Thousands of pixels with only dozens of labeled images cannot produce a full rank within class scatter.

Symptom

A direct solve fails or becomes extremely sensitive to small perturbations.

Regularization stabilizes the Fisher direction

Regularized scatter

$$S_W^{(\lambda)} = S_W + \lambda I, \qquad \lambda > 0$$

Direction

$$w_\lambda = (S_W + \lambda I)^{-1}(m_1 - m_2)$$

Effect

Small variance directions no longer receive arbitrarily large inverse weights.

Selection

Choose lambda with validation data.

Fisher direction in NumPy

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```
X1 = np.array([[1., 1.], [2., 0.], [2., 1.], [3., 0.]])
X2 = np.array([[1., 3.], [2., 2.], [2., 3.], [3., 2.]])
m1, m2 = X1.mean(axis=0), X2.mean(axis=0)

C1, C2 = X1 - m1, X2 - m2
Sw = C1.T @ C1 + C2.T @ C2
w = np.linalg.solve(Sw + 1e-6*np.eye(2), m1 - m2)
w = w / np.linalg.norm(w)

z1, z2 = X1 @ w, X2 @ w
threshold = 0.5 * (z1.mean() + z2.mean())
print(m1, m2, Sw, w, threshold, sep='\n')
```

Expected direction

The normalized vector points almost straight down.

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Fisher model checks

1. Compare training and validation separation on the projected axis.
2. Inspect projected distributions rather than only their means.
3. Report the threshold, priors, and loss assumptions separately.
4. Stress test the direction after small data and scaling changes.

A high criterion is not enough

A few outliers or a shifted deployment population can still produce poor decisions.

One problem, four boundary designs

Design	What is learned	Best first question
Linear	one score $w^\top x + w_0$	Can one straight boundary work?
Feature map	a linear score in $\phi(x)$	Which transformation exposes the pattern?
Piecewise linear	several local linear scores	Does the boundary need several faces?
Fisher	a projection with low within class spread	Which direction separates compactly?

Do not begin with complexity

Begin with the simplest geometry that can express the decision.

Integrated case: two tomato varieties

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Observed difficulty

Size separates most samples, but two growing regions create a bent boundary in color and firmness.

1. Plot the two features and try one linear score as a baseline.
2. Add interpretable transformed features if the bend has a known form.
3. Use local linear pieces if different growing regions form separate subgroups.
4. Use Fisher projection when a compact one dimensional inspection score is required.



Operational decision

Choose the threshold from the cost of sending a wrong variety to packaging.

From geometry to learning algorithms

This lecture

We designed decision functions and understood their geometry.

Next lecture

The perceptron will learn a separating direction from mistakes; decision trees will partition the space by questions.

Same data, different inductive bias

A learning rule is not only an optimizer. It also determines which boundaries are easy to discover.

Exit ticket

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1. State the difference between a generative and a discriminative classifier in one sentence.
2. Explain the sign and magnitude of a linear discriminant score.
3. Give one reason to use a feature map and one reason to use linear pieces.
4. Explain the numerator and denominator of the Fisher criterion.
5. For XOR, choose a model family and justify the choice.

A strong answer

Names the geometry, the information used, and one failure mode rather than listing formulas.